Machine Learning:
Reinforcement Learning & Review

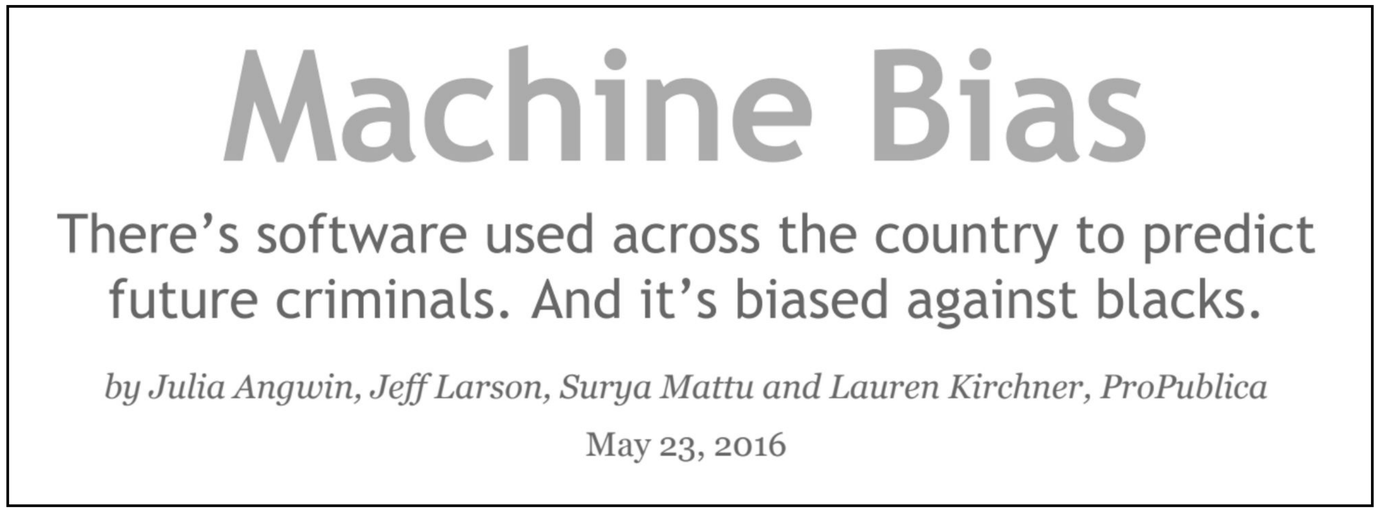
Brendan Tomoschuk

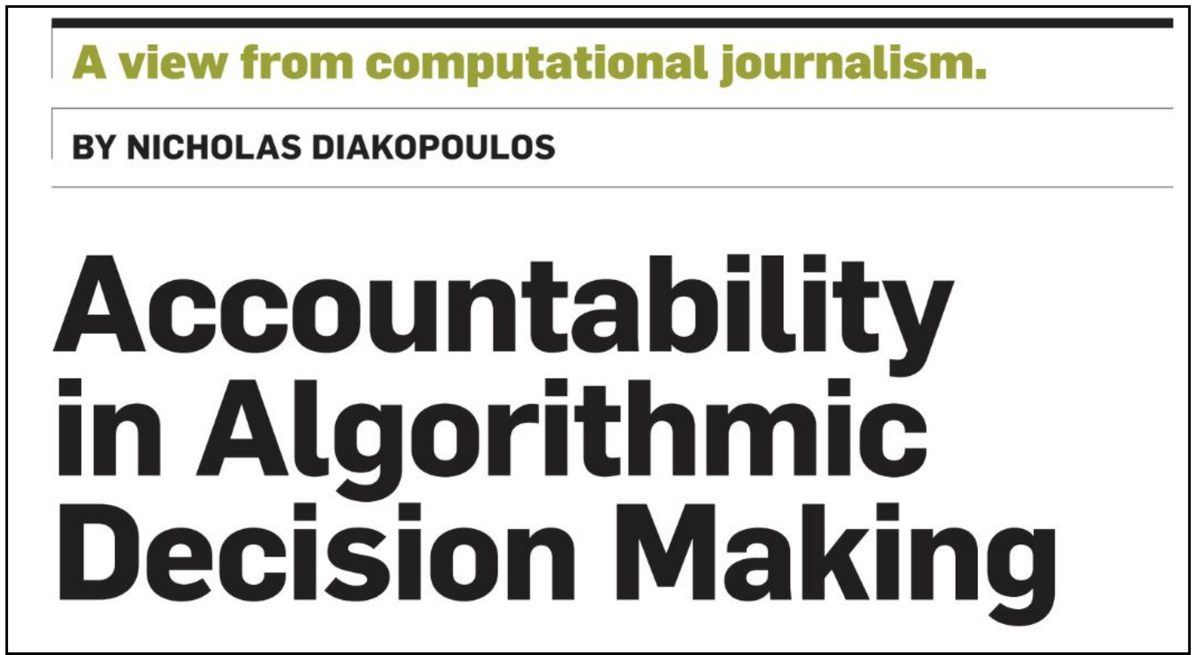
Reading(s) 5: *Machine Learning*

Theme: *Algorithms shape real lives*

What does it mean for an algorithm to be fair and unbiased?

Case study of a biased criminal risk-prediction algorithm





What is **algorithmic accountability?**

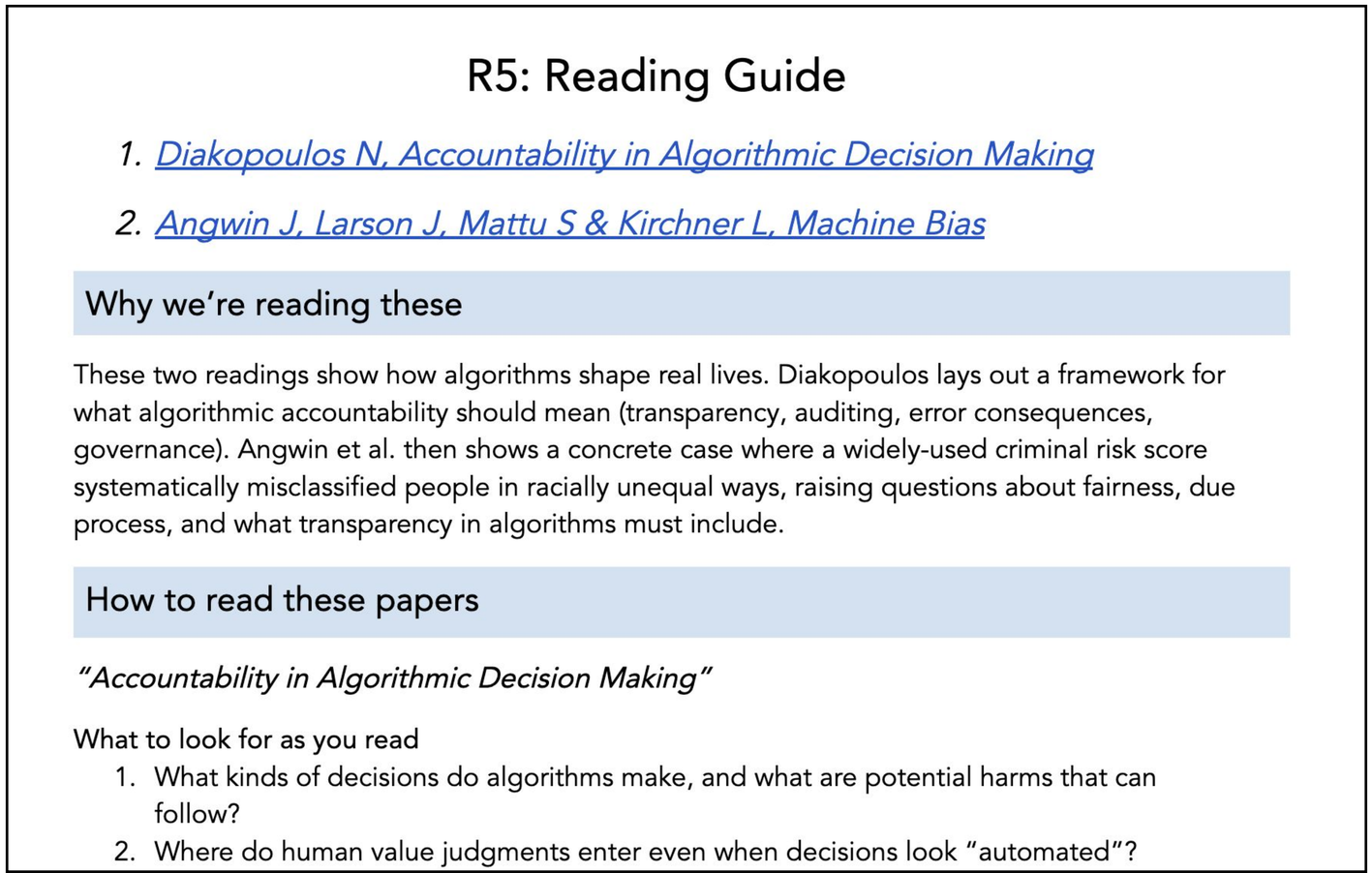
How do algorithms often make decisions on our behalf?

How should algorithms be made transparent to the public?

Reading(s) 5: *Machine Learning*

- Use **Reading Guide 5** to focus on the key themes and main ideas
  - Remember, R4 & R5 papers will be on Exam 2!

- **Reading 5 Quiz:**
  Due Wednesday Week 9
  - 10 MC questions
  - You have 60 mins and 2 attempts to complete it!
  - You may use the reading + reading guide + any notes!



R5: Reading Guide

1. *Diakopoulos N, Accountability in Algorithmic Decision Making*
2. *Angwin J, Larson J, Mattu S & Kirchner L, Machine Bias*

Why we're reading these

These two readings show how algorithms shape real lives. Diakopoulos lays out a framework for what algorithmic accountability should mean (transparency, auditing, error consequences, governance). Angwin et al. then shows a concrete case where a widely-used criminal risk score systematically misclassified people in racially unequal ways, raising questions about fairness, due process, and what transparency in algorithms must include.

How to read these papers

"Accountability in Algorithmic Decision Making"

What to look for as you read

1. What kinds of decisions do algorithms make, and what are potential harms that can follow?
2. Where do human value judgments enter even when decisions look "automated"?

Updates to class schedule



Week 7

- **Monday:** Machine Learning
- **Wednesday**: NO CLASS (Veteran's day)
- **Friday:** Language Models

Week 8

- **I am traveling so there will be no new material**
- **Monday:** [remote, recorded] Interview with Dr. Jolene Mork from Microsoft AI
- **Wednesday:** Andrew Cronan - Career Decision Making
- **Friday:** Class is CANCELLED

Midterm 2 — Monday November 23

- **Format:** 35 multiple-choice questions, 50 mins, ***CLOSED-NOTE***
  - ***Questions taken from***
    - ***lectures from “Data visualization” to “Language Models”***
    - ***Readings 4-5***
- **Bring:**
  - A pen or pencil
  - No need to bring a scantron
- **Reminders:**
  - Come early and get a seat to maximize everyone’s time!
  - Students with accommodations will take the exam at the Triton Testing Center.

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course code:

WSTAPX

You want to visualize some data that has a lot of features (high-dimensional). Which kind of model should you use?

(A) Neural networks

(B) Linear regression

(C) Principal components analysis (PCA)

(D) K-means clustering

(E) I'm not sure / I'm confused

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Reinforcement Learning

Learning from trial-and-error

Machine Learning

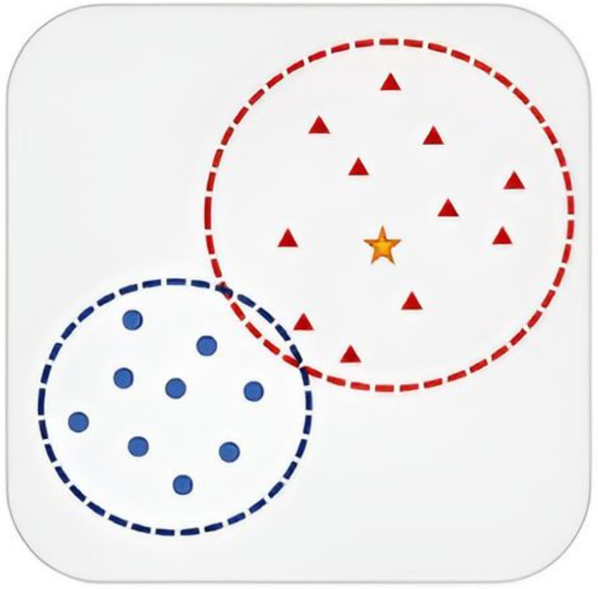
Learning from data to predict or generalize to unseen data

Learning from trial-and-error interactions with the environment

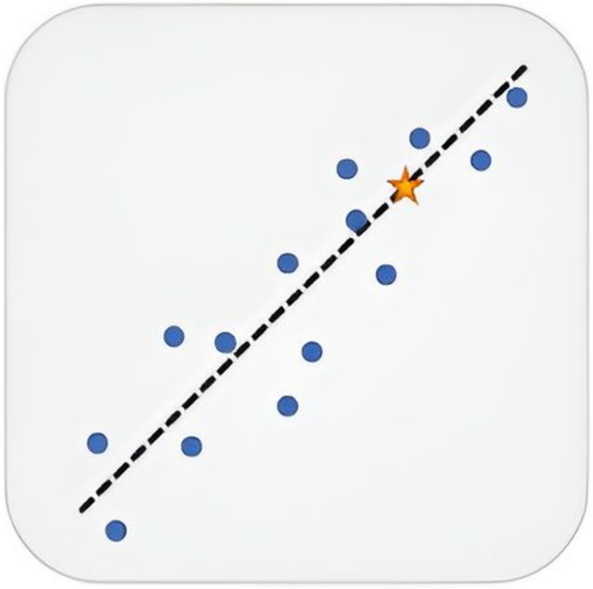
Supervised Learning

Model training with labelled data

Classification



Regression

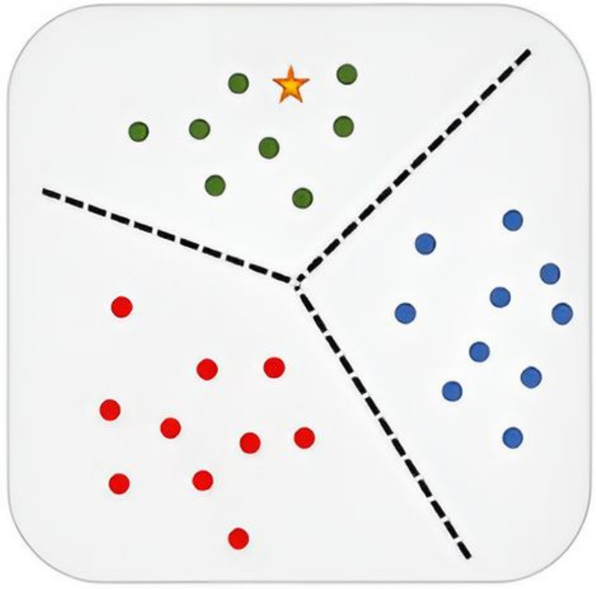


Unsupervised Learning

Model training with unlabelled data

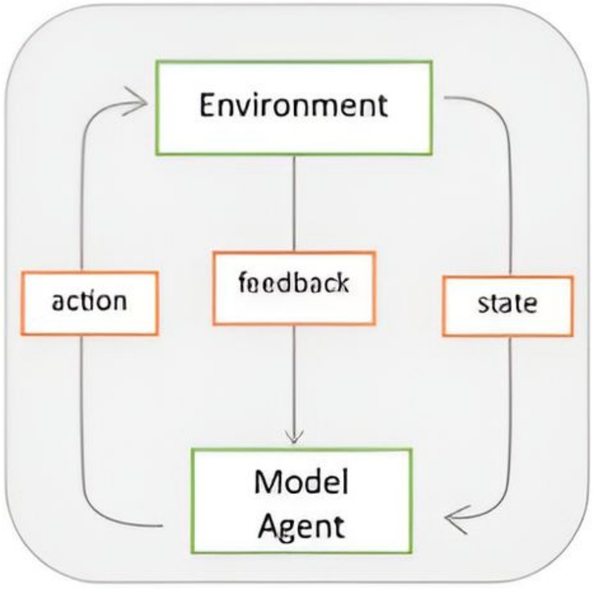
Discovering patterns in data

Clustering



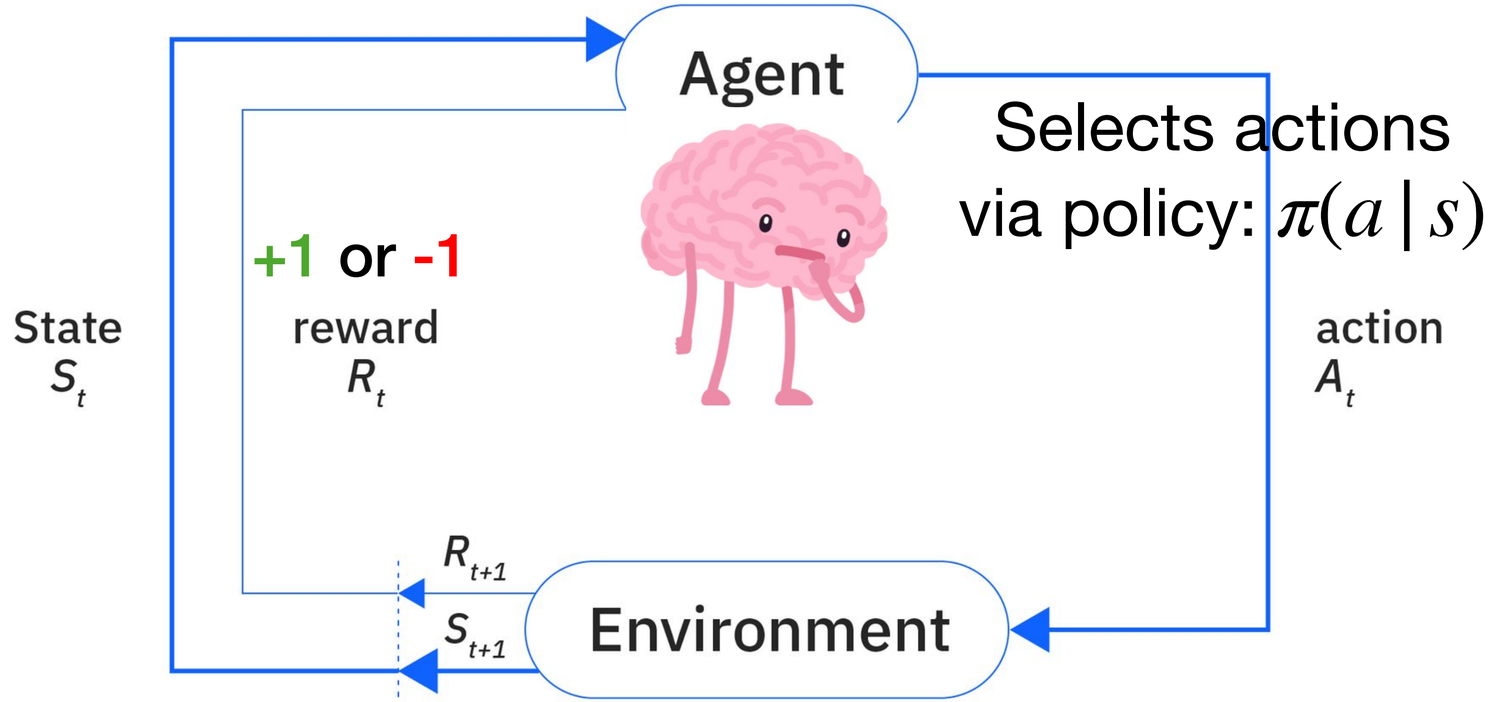
Reinforcement Learning

Model take actions in the environment then received state updates and feedbacks



Reinforcement Learning

Reinforcement learning (RL): models that describe how agents learn from trial and error interactions in their environment



Agent's goal:
Maximize long-term reward

Learning a policy $\pi(a \mid s)$

Assume: The chef changes every day, so food is not consistent

State: weekday	Monday	Tuesday	Wednesday	Thursday	Friday
Action: food	?	?	?	?	?



Goal: figure out where you should eat each day to maximize overall reward

(find best place to eat on each day)

Learning a policy $\pi(a \mid s)$

Goal: learn a policy $\pi(a \mid s)$ that maximizes overall reward

State: weekday

Monday	Tuesday	Wednesday	Thursday	Friday

Action: food

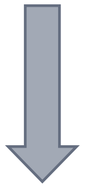


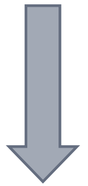











Reward: how good the food was

+0	+1	+0	+1	+0

Each week, you try different actions in each state (trial-and-error)

Learning a policy $\pi(a \mid s)$

Goal: learn a policy $\pi(a \mid s)$ that maximizes overall reward

State: weekday

Monday	Tuesday	Wednesday	Thursday	Friday

Action: food



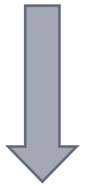


















Reward: how good the food was











+1 +1 +1 +1 +1

…until you find the "optimal" policy (maximizes reward)

Learning a policy $\pi(a \mid s)$

Goal: learn a policy $\pi(a \mid s)$ that maximizes overall reward

State: weekday

Monday	Tuesday	Wednesday	Thursday	Friday

Action: food



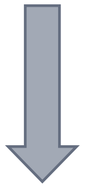




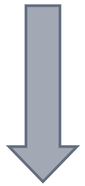














Reward: how good the food was









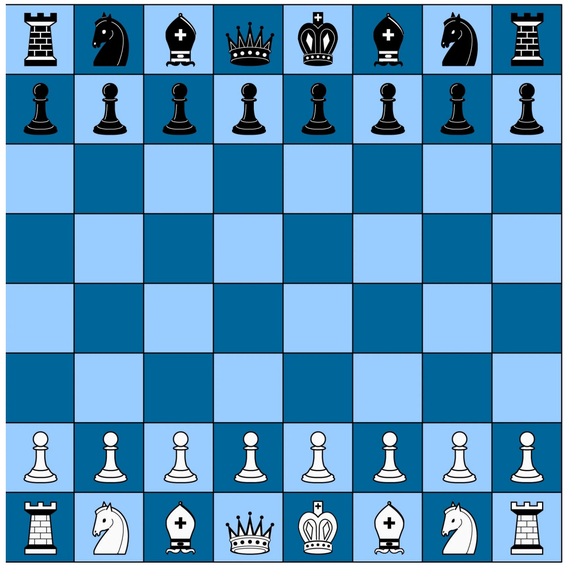


+1 +1 +1 +1 +1

Takeaway: do things that lead to rewarding experiences, avoid things that don't!

Reinforcement Learning

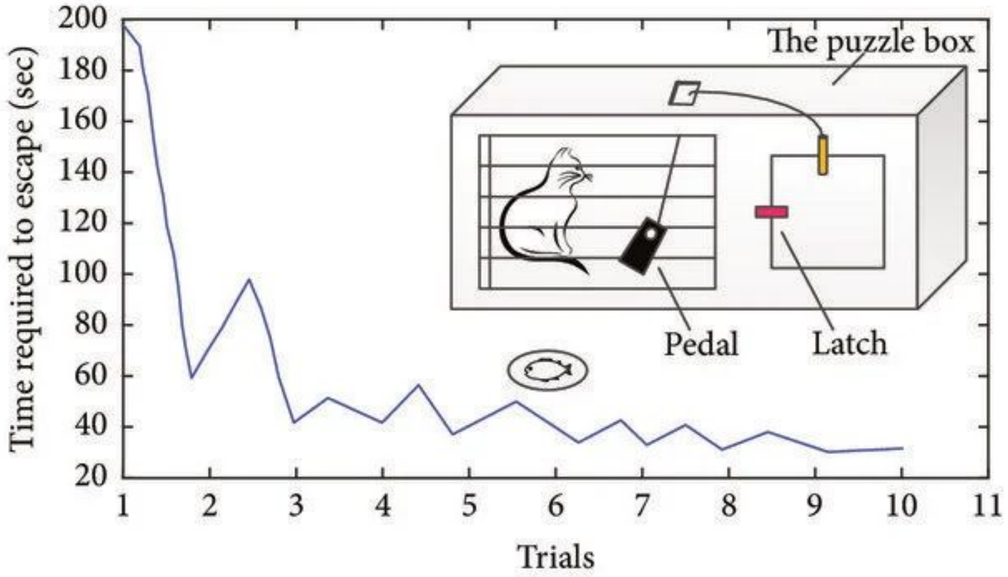
- Reinforcement learning algorithms closely mirror how humans (and animals) learn from the real world:



Learning the best chess plays



Learning social norms from social interactions



Cats learning to escape from a box

- So if we want to build AI that can interact with the world as humans do, RL seems like a good place to start!

Training AI to play human games using RL

Atari
(Deep Q-network)



2015

Go
(AlphaGo)



2016

Chess
(AlphaZero)

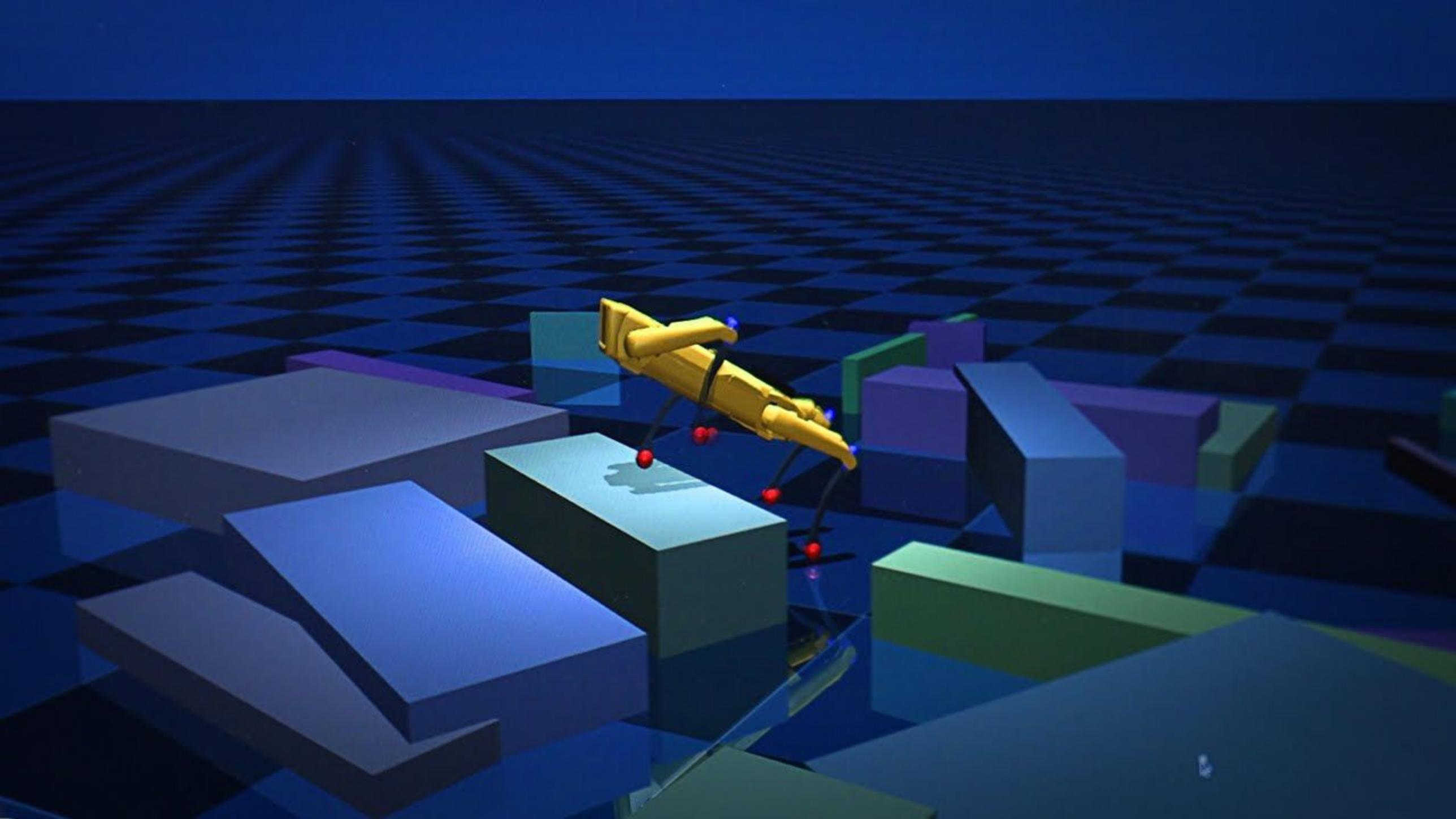


2018

StarCraft
(AlphaStar)



2019







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course code:

WSTAPX

Netflix wants to group users into different types based on their TV show preferences, but does not know in advance what the groups should be. Which branch of machine learning is this?

(A) Supervised Learning

(B) Unsupervised Learning

(C) Reinforcement Learning

(D) I'm not sure / confused

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How do I study?

1. I will test you on **which visual / EDA / statistical analysis / ML model** to use in any given scenario
2. **Questions to ask yourself when choosing a [visualization, EDA method, statistical analysis, ML model]:**
   - *Am I visualizing / exploring / analyzing / modeling* [one, two, or multiple] *variable(s)*?
   - *Is my variable(s)* quantitative or qualitative (categorical)?
   - *Which visualization / EDA / statistical analysis / ML model is suitable for* [one, two, or multiple] [quantitative or qualitative (categorical)] *variable(s)*?
3. ***Go to Office Hours*!**
4. ***Go to discussion section!* (Exam 2 review)**

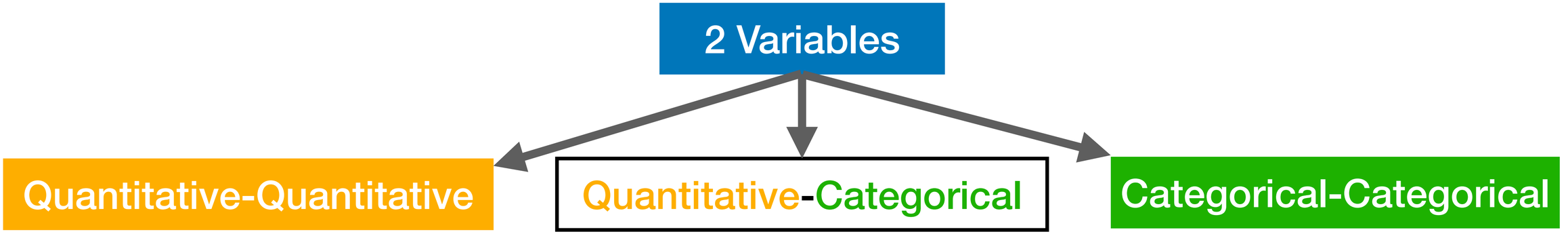
Study sheet: Types of EDA and their visualizations

- **Univariate EDA:** understanding each individual variable by itself (mean/median/std)
  - Quantitative: understand summary stats: mean/median, std, etc. and distribution, skew, outliers
    - ***Visualization:*** box plots, histograms, density plot
  - Qualitative: summarize categorical counts for nominal or ordinal data
    - ***Visualization:*** bar plots, pie charts, frequency tables
- **Bivariate EDA:** draw conclusions about the relationship between two variables (comparison or association)
  - Quantitative ↔ quantitative: test association in values
    - ***Visualization:*** scatterplots!
  - Categorical ↔ quantitative: examine difference in means or distribution across groups
    - ***Visualization:*** side-by-side box plots (distribution), bar plots (means)
  - Categorical ↔ categorical: association or difference in proportions
    - ***Visualization:*** contingency tables
- **Multivariate EDA:** drawing conclusions about relationships between variables while accounting for other variables; detect confounding and interaction effects
    - ***Visualization:*** scatter plot colored by different categories

Study sheet: Types of statistical inference/analysis

- **Univariate inference:** estimate a single population parameter (e.g., μ, σ) about a single variable
  - ***Inference tools:*** standard error, confidence intervals
- **Bivariate inference:** draw conclusions about the relationship between two variables (comparison or association)
  - Quantitative ↔ quantitative: test association in values
    - ***Inference tools:*** correlation, slope
  - Categorical ↔ quantitative: difference in means across groups
    - ***Inference tools:*** t-test, ANOVA
  - Categorical ↔ categorical: test association or difference in proportions
    - ***Inference tools:*** chi-square test
- **Multivariate inference:** drawing conclusions about relationships between variables while accounting for other variables; detect confounding and interaction effects *(won't be tested)*

Make a study flow chart (example)



- viz/EDA: *scatterplot*
- stats: *Pearson's correlation*

- viz/EDA: *contingency table*
- stats: *t-test*

- viz/EDA: *contingency table*
- stats: *Chi-squared test*

Do the same for univariate & multivariate analysis & ML models!

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course code:
WSTAPX

You want to predict what your letter grade in this class is going to be (A, B, C, D, F) from your exam 1 score. *What kind of problem is this, and what kind of model would you use?*

Ask yourself: Outcome variable(s) quantitative or qualitative?

(A) Regression; simple linear regression

(B) Classification; multiple linear regression

(C) Regression, polynomial regression

(D) Classification, logistic regression

(E) I'm not sure / I'm confused

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course code:
WSTAPX

You want to analyze the differences between the mean exam grades of COGS 9 (Fall 2026) vs. COGS 9 (Fall 2025).

Which statistical inference test would you use?

(A) Chi-squared test

(B) t-test

(C) Linear regression

(D) Pearson's correlation

(E) I'm not sure / I'm confused

Ask yourself:

1. [One, two, multiple] *variable(s)*?
2. *Variable(s)* quantitative or qualitative?

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Which kind of visualization would you use?

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course code:

WSTAPX

You want to test the association between temperature and humility in San Diego.

Which combination of EDA/visualization, and statistical analysis would you use?

(**A**) Scatterplot, Chi-squared test

(**B**) Bar plot, t-test

(**C**) Scatterplot, Pearson's correlation

(**D**) Bar plot, Pearson's correlation

(**E**) I'm not sure / I'm confused

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Discuss (~5 minutes)

Your data science question is: Does Zip code, atmospheric pressure and temperature predict whether or not it will rain (true/false)?

1. What EDA(s) do you need to consider?
2. What model would you run?
3. Which evaluation metric would you use for your model?

Study Sheet: Machine Learning Models

Supervised Learning: Predict Y from X

Regression: Predicting quantitative Y

- Linear regression (simple & multiple)
- Non-linear regression

Classification: Predicting qualitative (categorical) Y

- Logistic regression

Both regression & classification:

- Decision trees
- Neural networks

Unsupervised Learning: Discovering structure in data

Dimensionality reduction: Summarizing or visualizing high dimensional data

- Principal components analysis (PCA)

Clustering: Grouping the data by similarity of features

- K-means clustering
- Hierarchical clustering

Reinforcement Learning: Learning from trial-and-error in the environment

Takeaway: Choose an appropriate model depending on your modeling **goal**!